

Cortical beta oscillations map to shared brain networks modulated by dopamine

Reviewed Preprint

v2 • October 28, 2024

Revised by authors

Reviewed Preprint

v1 • May 17, 2024

Meera Chikermane, Liz Weerdmeester, Nanditha Rajamani, Richard M Köhler, Timon Merk, Jojo Vanhoecke, Andreas Horn, Wolf-Julian Neumann 

Movement Disorders and Neuromodulation Unit, Department of Neurology, Charité – Universitätsmedizin Berlin, Berlin, Germany • Center for Brain Circuit Therapeutics, Department of Neurology, Brigham and Women's Hospital, Boston, USA • Departments of Neurology and Neurosurgery, Massachusetts General Hospital, Boston, USA • Einstein Center for Neurosciences Berlin, Humboldt Universität, Berlin, Germany

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study poses an **important** step forward in understanding the brain-network embedding of beta oscillations. The study advances our circuit-level understanding of the pathophysiology associated with dopaminergic alterations in psychiatric or neurological disorders. The study provides **compelling** evidence that beta oscillations across the neocortex and basal ganglia map onto shared functional and structural networks that show significant positive correlations with dopamine receptors.

<https://doi.org/10.7554/eLife.97184.2.sa4>

Abstract

Brain rhythms can facilitate neural communication for the maintenance of brain function. Beta rhythms (13–35 Hz) have been proposed to serve multiple domains of human ability, including motor control, cognition, memory and emotion, but the overarching organisational principles remain unknown. To uncover the circuit architecture of beta oscillations, we leverage normative brain data, analysing over 30 hours of invasive brain signals from 1772 channels from cortical areas in epilepsy patients, to demonstrate that beta is the most distributed cortical brain rhythm. Next, we identify a shared brain network from beta dominant areas with deeper brain structures, like the basal ganglia, by mapping parametrised oscillatory peaks to whole-brain functional and structural MRI connectomes. Finally, we show that these networks share significant overlap with dopamine uptake as indicated by positron emission tomography. Our study suggests that beta oscillations emerge in cortico-subcortical brain networks that are modulated by dopamine. It provides the foundation for a unifying circuit-based conceptualisation of the functional role of beta activity beyond the motor domain and may inspire an extended investigation of beta activity as a feedback signal for closed-loop neurotherapies for dopaminergic disorders.

Introduction

Oscillatory neural activity is proposed to orchestrate brain function, by providing multilateral rhythmic synchronisation of neural excitability and action potential firing¹. One prominent example in the human brain is the beta rhythm (13 – 35 Hz), which has traditionally been described as a phenomenon of the motor domain, primarily localised to motor cortex and important for the pathophysiology of movement disorders, such as Parkinson's disease (PD)^{2,3}. In PD, beta activity, especially in deeper brain structures, such as the basal ganglia, has been associated with the loss of innervation with dopamine⁴, a neurotransmitter critically involved in synaptic plasticity and reinforcement learning at cortico-striatal synapses⁵. This relationship is mirrored in rodent and non-human primate models of PD, corroborating the role of dopamine as a key neurotransmitter governing beta synchrony. Meanwhile, the putative function of beta modulation in both human and animal literature has been extended within and beyond PD, to diverse domains of cognition³, memory⁶ and emotions⁷, but only few studies so far have sought to relate these observations to dopamine signalling. For each of these domains, localised synchronisation or desynchronisation phenomena in the beta frequency range have been described in specific brain regions. What the common denominators of these brain regions are, and whether beta activity has a general structural and functional overlap with dopaminergic innervation remains unknown. This reflects a significant roadblock to a fundamental understanding of the circuit architecture of beta activity. To overcome this roadblock, we have developed a unique multimodal research data pipeline combining normative sources of invasive brain signal recordings, MRI connectomics and positron emission tomography (PET). We address the hypothesis that beta activity is a ubiquitous synchronisation phenomenon, that is characterised by whole-brain networks with subcortical nuclei that are under modulatory control of dopamine.

Results

Beta activity is a widely distributed resting brain rhythm in invasive cortical recordings

First, we aimed to identify the spatial distribution of cortical sources of beta activity in the resting human brain. Electrophysiology studies describing brain oscillations often rely on non-invasive electroencephalography (EEG) or magnetoencephalography (MEG) that provide access to brain signal recordings in healthy subjects. While beamforming and other source reconstruction techniques can provide estimates of the spatial peaks of certain brain activity patterns in these recordings, it can be hard to rule out that the derived brain regions are free from influence of volume conduction and current mixing from neighbouring brain areas⁸. Thus, estimates of beta activity in prefrontal areas identified with EEG or MEG could to some extent, be the result of motor cortex activity, contaminating sensors in frontal locations. Invasive recordings directly from brain tissue can circumvent such contamination, but are unavailable in healthy subjects. Here, we analysed over 30 hours of invasive resting state brain signals from an open atlas⁹ of the normal electroencephalogram in 1772 channels of either electrocorticography (ECoG; n=258) or stereoelectroencephalography (sEEG; n=1514) covering the human cortex. The atlas is derived from 106 epilepsy patients undergoing monitoring for seizure activity for respective surgery, but the signals were selected based on strict criteria to ensure their origin from healthy brain regions, in relevant distance to epileptogenic zones or lesions⁹. The localised recordings from bipolar channels of one minute resting periods with eyes closed were taken to the frequency domain using Morlet wavelets and parametrised using the “fitting oscillations and one over f” (FOOOF) toolbox¹⁰, to extract the maximum peak amplitude of the periodic component of beta activity

(13 – 30 Hz). For comparison and as a control, we extended this analysis to other canonical brain rhythms, including theta (4 – 8 Hz), alpha (8 – 12 Hz) and gamma (30 – 100 Hz) frequency bands. The frequency band with the highest peak amplitude was identified using the extracted peak parameter (pw) for each channel and depicted as the dominant rhythm for the respective localisation (**Figure 1**). In case of multiple peaks within the same frequency band, we focused the analysis on the peak with the highest amplitude. This revealed that 15.6% (277) channels were identified as theta dominant, 22.4% (397) as alpha dominant, 56.6% (1005) as beta dominant and 5.2% (93) as gamma dominant channels. The focus of this article is beta oscillations, so all further analyses will refer to beta with alpha for comparison as it was the second most frequently occurring frequency band after beta. It should be noted that that alpha recordings were performed in eyes closed which is known to increase alpha power, which may influence the generalisability of the alpha maps to an eyes open condition. However, given that our primary use of alpha was to act as a control, we believe that this should not affect the interpretability of the key findings of our study. Further information on the topography of theta channels are shown in supplementary figure 1.

To investigate the regional distributions of frequency bands, we counted the occurrence of each dominant rhythm in gyri and lobes (see **Figure 1** and **Table 1**), by projecting the coordinates in MNI (Montreal Neurological Institute) space to the automated anatomical labelling (AAL) atlas parcellation¹¹. Notably, many channels showed more than one oscillatory peak, and all covered lobes had channels with dominant rhythms in each, theta, alpha and beta frequency bands (see **Table 1**). Beta activity was the most frequent and most widely distributed resting rhythm across all lobes of the human brain including frontal, temporal, cingulate, insula and parietal lobes. Interestingly, the occipital lobe was the only brain region, that showed higher percentage of alpha dominant channels (53/94; 56.4%). The coverage of subdural ECoG grids or tissue penetrating sEEG electrodes could vary, but within overlapping regions, no major differences in the distribution of brain rhythms were observed for electrode type (see supplementary Table 1).

MRI connectomics reveal shared cortico-subcortical beta networks

Oscillatory synchronisation across brain regions may shape neural communication through modulation of synaptic sum potentials and neural firing¹. Oscillatory connectivity can be computed through correlation and modelling of phase and amplitude relationships between brain signals from multielectrode recordings. The spatial coverage of such computations, however, is inherently limited to brain regions that are accessible with the respective recording technique. For non-invasive M/EEG, the analysis of connectivity is limited to superficial cortical brain areas, and can be biased by noise and volume conduction¹². Invasive recordings in both human and animal studies provide more robust estimates of oscillatory connectivity but are limited to a selected number of channels from defined brain regions¹³. Here, we aimed to investigate the whole brain circuit representation of beta activity, which is impossible with non-invasive neurophysiology approaches. Instead, we tested the hypothesis that cortical brain areas that exhibit resting beta share a global brain network by means of MRI connectomics. For this, we derived the methodology from lesion and DBS network mapping¹⁴, which has provided groundbreaking insights into the circuit nature of brain lesion and neurostimulation effects and applied it to reveal the shared network fingerprints of brain regions that exhibit a neurophysiological phenomenon; namely, dominant beta activity. For this we followed a multimodal approach, investigating neurophysiologically identified brain regions by means of MRI connectomics, generating functional and structural connectivity maps for each brain signal recording location, as previously described¹⁵. In brief, the MNI coordinates of the invasive electrode locations were used as seeds to compute whole-brain connectivity maps based on MRI data from 1000 healthy subjects as available through the Brain Genomics Superstruct project, and MRI data from 32 subjects as available through the Human Connectome Project (HCP) as part of the Lead Mapper pipeline¹⁶. While the connectivity maps estimated with functional MRI (fMRI) are based on the covariance blood oxygenation level dependent signals (BOLD) that provide an

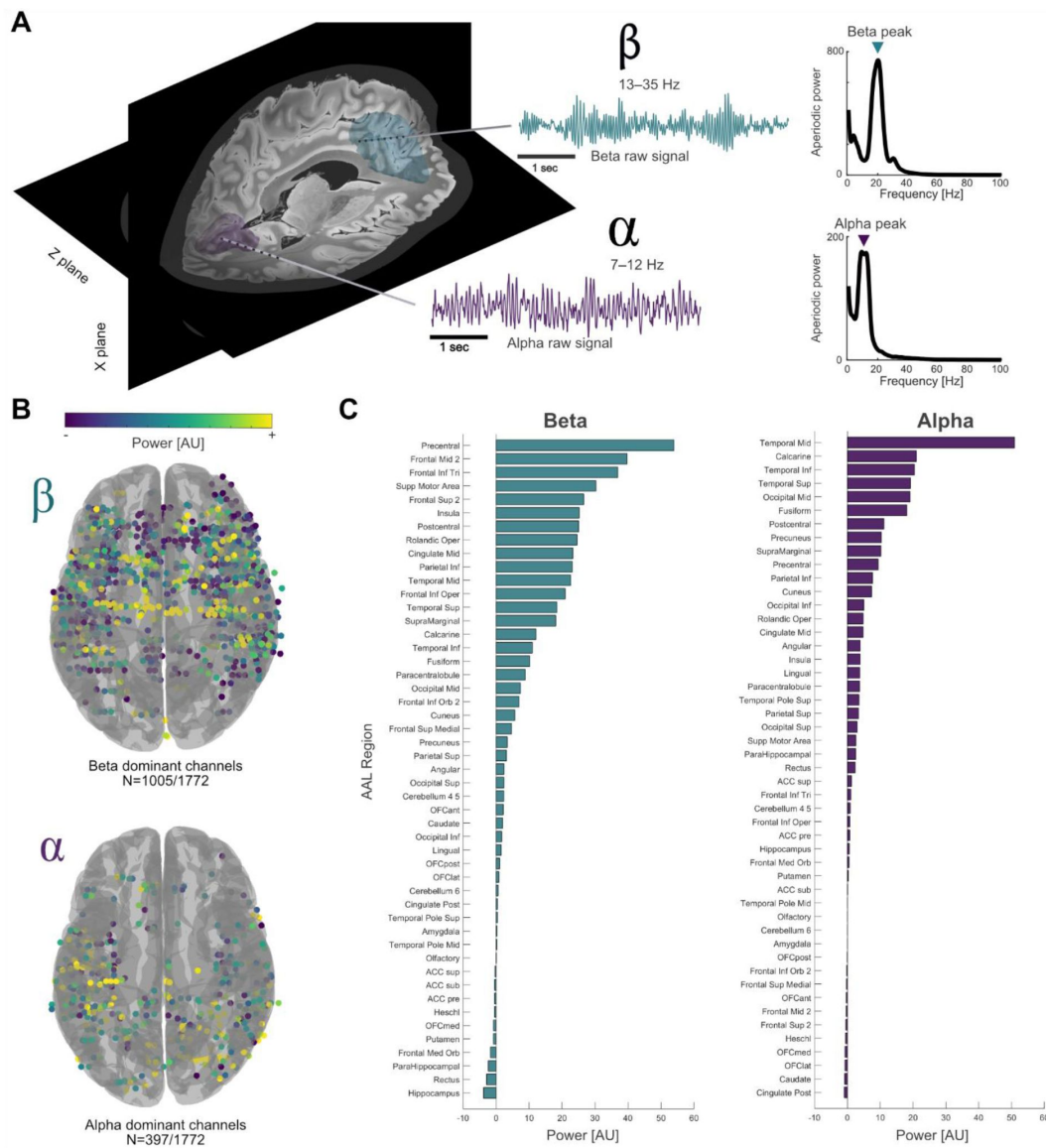


Figure 1

Beta activity is a widely distributed resting brain rhythm in invasive cortical recordings.

(A) Exemplar sEEG electrodes with dominant beta (teal; 20 Hz) or dominant alpha (purple; 9 Hz) rhythms, alongside the respective raw signal and parametrised power spectrum. (B) 1005/1772 channels showed higher maximum peak power in beta (13 – 30 Hz) than theta (4 – 8 Hz), alpha (8 – 12 Hz) and gamma (30 – 100 Hz). 397/1772 were alpha dominant (theta and gamma not shown). (C) Beta dominant channels were distributed across the entire cortex, from occipital to frontal regions, with highest density in sensorimotor areas like precentral gyrus, frontal middle gyrus, frontal inferior gyrus and supplementary motor area. Alpha channels were largely concentrated posteriorly in temporoparietal and occipital areas based on the automatic anatomical labelling atlas¹¹. Neither alpha nor beta activity showed systematic hemispheric differences ($p > 0.05$).

Lobe	Channels/ lobe	ECoG	sEEG	Sum theta channels		Sum alpha channels		Sum beta channels	
Frontal	616	122	494	73	11.80%	38	6.20%	458	73.40%
Temporal	519	68	451	106	20.40%	180	35.00%	216	41.60%
Cingulate	116	0	116	12	10.30%	21	18.10%	71	61.20%
Insula	128	2	126	19	14.80%	21	16.40%	86	67.20%
Amygdala	3	0	3	1	33.30%	0	0%	1	33.30%
Parietal	272	52	220	50	18.40%	73	26.80%	139	51.10%
Occipital	107	13	94	15	14.00%	62	57.90%	27	25.20%
Subcortex	11	1	10	1	9.10%	2	18.20%	7	63.60%

Table 1

Spread of invasive electrophysiological channels.

Given the low number of channels exhibiting resting state gamma activity, we excluded those from the present table.

estimate of functional coactivation, the structural connectivity maps are based on diffusion weighted MRI (dMRI) and provide a relative quantification of major anatomical white matter tracts. To allow for a statistical comparison of connectivity patterns, we compared voxel-wise connectivity estimates from recording locations with dominant beta rhythms, to recording locations with dominant alpha rhythms using mass univariate two-sample t-tests in Statistical Parametric Mapping software, with a family wise error corrected voxel threshold set to $p < 0.05$.

Our analysis revealed a shared whole brain network for cortical brain regions that exhibit dominant beta oscillations, with marked and highly significant differences (**Figure 2**) to the control network for alpha oscillations, both in whole-brain functional and structural connectivity patterns (see Supplementary Figure 2 for unthresholded averages). Most prominently, the beta network showed significantly higher positive connectivity to prefrontal, cingulate and parietal cortical areas and the ventral thalamus and basal ganglia, including substantia nigra, striatum, pallidum and subthalamic nucleus. In contrast, whole brain networks seeded from alpha-dominant channels showed a more widespread distribution across the posterior cortical areas, occipital and temporal lobes in comparison to the beta networks. All connectivity values including their sign are shown in figures as brain region averages parcellated with the automatic anatomical labelling atlas in supplementary figures 3 and 4.

The connectomic beta network correlates with molecular markers of dopamine signalling

Invasive brain signal recordings in patients undergoing deep brain stimulation (DBS) for Parkinson's disease (PD) have robustly established a strong interaction between dopaminergic tone and amplitude of beta oscillations at rest, e.g. by measuring changes in activity after withdrawal and administration of dopaminergic agents¹⁷. However, this interaction has been largely neglected outside the field of PD, likely because research methods to establish direct relationships between beta and dopamine in healthy subjects are not readily available. With the present project, we aimed to characterise the circuit architecture of cortical beta oscillations. We found that cortical brain areas, that exhibit resting beta activity share a common brain network. This network, unrelated to pathology, includes strong connectivity with the basal ganglia, the primary target for dopaminergic innervation from the substantia nigra and the ventral tegmental area in the human brain. To further quantify the potential relationship between the beta network and dopamine signalling, we made use of another openly available human neuroscience resource: a database of positron emission tomography (PET) imaging, a molecular imaging technique that relies on intravenous injection of a radioactively labelled molecular tracer, that can be visualised to indicate brain areas with high uptake of neurotransmitters or related molecules in the brain¹⁸. Specifically, we analysed to what degree the spatial uptake patterns of dopamine, as measurable with fluorodopa (FDOPA; cohort average of 12 healthy subjects) and other dopamine signalling related tracers that bind D1/D2 receptors (average of N=17/44 respectively healthy subjects) or the dopamine transporter (DAT; cohort average of N=180 healthy subjects) were correlated with the unthresholded MRI connectivity maps. To reduce the dimensionality and complexity of this analysis, we first applied spatial normalisation and then averaged the abovementioned cohort images to yield a PET based dopamine aggregate map (**Figure 3**). Next, we resliced the beta network map and the PET images to allow for a meaningful comparison, using a combined parcellation with 476 brain regions that include cortex¹⁹, basal ganglia²⁰, and cerebellum²¹. Here, each parcel – which was a collection of voxels belonging to a particular brain region – from the connectivity map was correlated with the same parcel containing average neurotransmitter uptake from the respective PET scan (see **Figure 3A**). In this way non-parametric Spearman's correlations between PET intensity and structural and functional connectivity maps of beta networks were obtained, which indicate to what degree the spatial distribution of connectivity is similar to the distribution of neurotransmitter uptake. This revealed a significant positive relationship between the beta functional and structural networks and

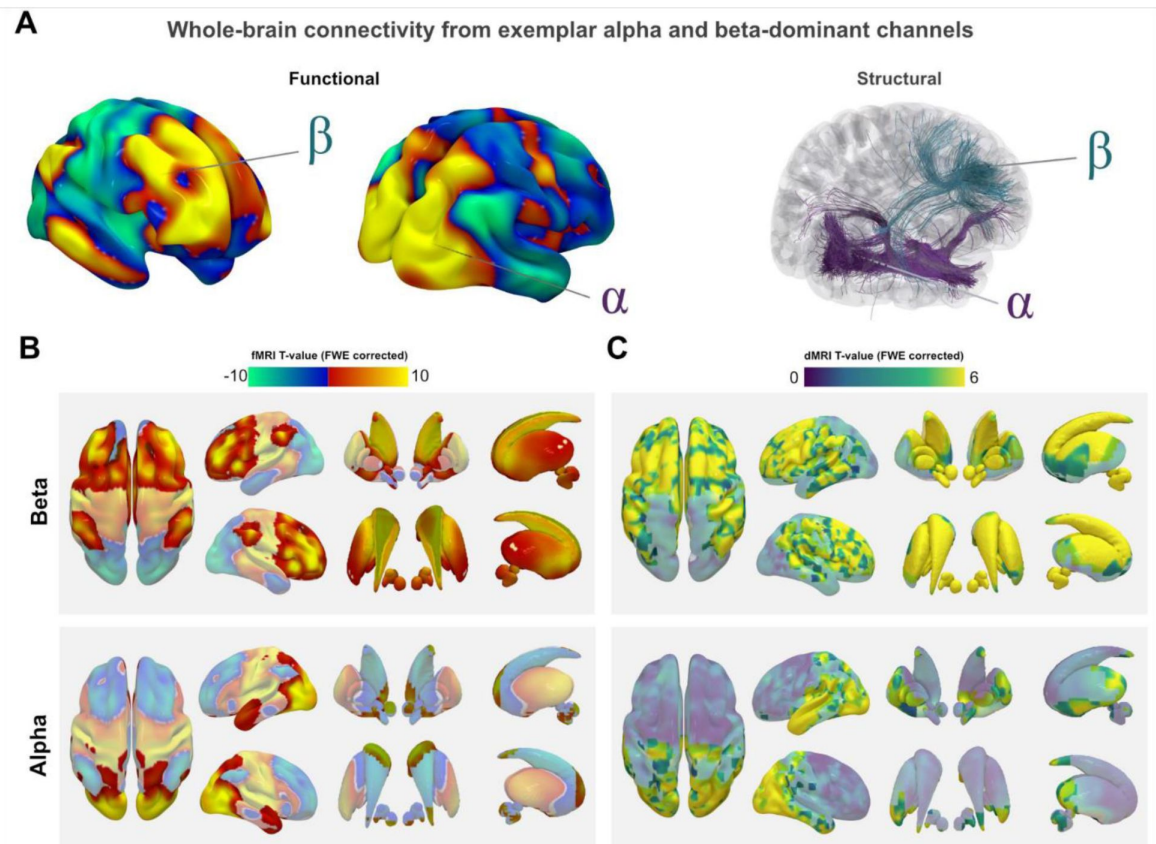


Figure 2

MRI connectomics reveal shared cortico-subcortical beta networks.

Individual network fingerprints (**A**) seeded from representative beta (top) and alpha (bottom) dominant electrodes highlight the methodological approach and showcase the distinct connectivity patterns that arise in dependence of the connected brain regions (for raw signal traces from these locations see [Figure 1](#)). Aggregated functional (**B**) and structural (**C**) connectivity maps across all beta (top) and alpha (bottom) dominant electrodes were subjected to mass univariate t-tests comparing beta vs. alpha networks with Statistical Parametric Mapping software and visualised as T-maps with significant clusters identified using family-wise error correction (shown opaque). Channel locations from beta dominant channels were associated with more robust functional and structural connectivity to frontal cortex and the basal ganglia, when compared to alpha channel locations, which were more connected to the occipital cortex.

dopamine uptake for cortex ($\rho=0.22/0.40$ for functional/structural networks, $P=0.0001$) and basal ganglia ($\rho=0.50/0.33$, $P=0.0001$), but not cerebellum (all $P>0.05$). These patterns were largely qualitatively reproduced, when the analysis was performed on the individual tracers (see Supplementary Figure 5). Additionally, we repeated the analysis with PET images of GABA uptake, to serve as an unspecific molecular control, which revealed no significant relationship between the beta functional network and GABA uptake cortically ($\rho=0.04$, $P=0.22$), in the basal ganglia ($\rho=0.07$, $P=0.27$) or cerebellum ($\rho=0.10$, $P=0.22$). The structural beta network revealed low correlation with GABA ($n=16$ healthy individuals) uptake for cortical areas ($\rho=0.17$, $P=0.0003$), negative correlations for basal ganglia ($\rho=-0.55$, $P=0.0001$) and no correlation for cerebellum ($P>0.05$). In summary, we found significant and specific overlap of the functional and structural connectivity maps seeded from beta-dominant cortical brain areas with the spatial patterns of dopamine uptake in the human brain.

Discussion

The present study combined three openly available datasets of invasive neurophysiology, MRI connectomics and molecular neuroimaging in human subjects to characterise the spatial distribution of brain regions exhibiting resting beta activity, their shared circuit architecture, and its correlation with molecular markers of dopamine signalling in the human brain. Our results can be broken down into three key insights: First, we demonstrate that the beta rhythm is the most widely distributed dominant neural synchronisation phenomenon, extending far beyond motor areas. Next, we show that these distributed areas share a common functional and structural whole-brain network that connects frontal, cingulate and other large-scale brain areas with the basal ganglia. Finally, we show that this network shares significant spatial overlap with molecular markers of dopamine signalling. We interpret these findings as supportive evidence for the hypothesis that beta activity is generated and sustained by a cortico-subcortical brain network that is modulated by dopamine. This conceptualisation moves away from functionally specific domains, towards a more holistic circuit understanding of beta activity as a brain circuit synchronisation phenomenon.

Brain functions associated with beta oscillations share brain networks modulated by dopamine

Our study investigates the connectivity profile of cortical brain regions that spontaneously exhibit beta oscillations as their dominant brain rhythm. Our results suggest that one common denominator of brain regions that generate beta activity, is their affiliation with a large-scale global brain network that is modulated by dopamine. However, we do not see this concept as contradictory to previous reports of functional correlates of beta activity in the domains of motor control, cognition, memory and emotion^{3,6,7,22}. Instead, we believe that the overlap in oscillatory activity across these domains can be conceptualised through a circuit definition of beta activity and its relationship to dopamine signalling. Foremost, in the motor domain there is direct evidence that pathological changes in dopamine causally modulates beta activity, as seen in human patients with Parkinson's disease¹⁷, but also in animals lesioned with neurotoxins targeting dopaminergic neurons, such as 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) in non-human primates²³ or 6-hydroxydopamine (6-OHDA) in rodents²⁴. Moreover, the robust beta modulation that is elicited by voluntary action in sensorimotor cortex is long known. In our study, motor and sensorimotor cortex showed higher beta peak amplitudes than any other brain regions. Thus, it is tempting to infer that beta activity and its relation to dopamine is characteristic to the motor domain. On the other hand, numerous studies have reported specific changes in beta activity in relation to many other neuroscientific domains that define human ability. For example, it was shown that beta is implicated in working memory²⁸, utilisation of salient sensory cues²⁹, language processing³⁰, motivation³¹, sleep³², emotion recognition³³, mood³⁴ and may even serve as a biomarker for depressive symptom severity in the anterior cingulate

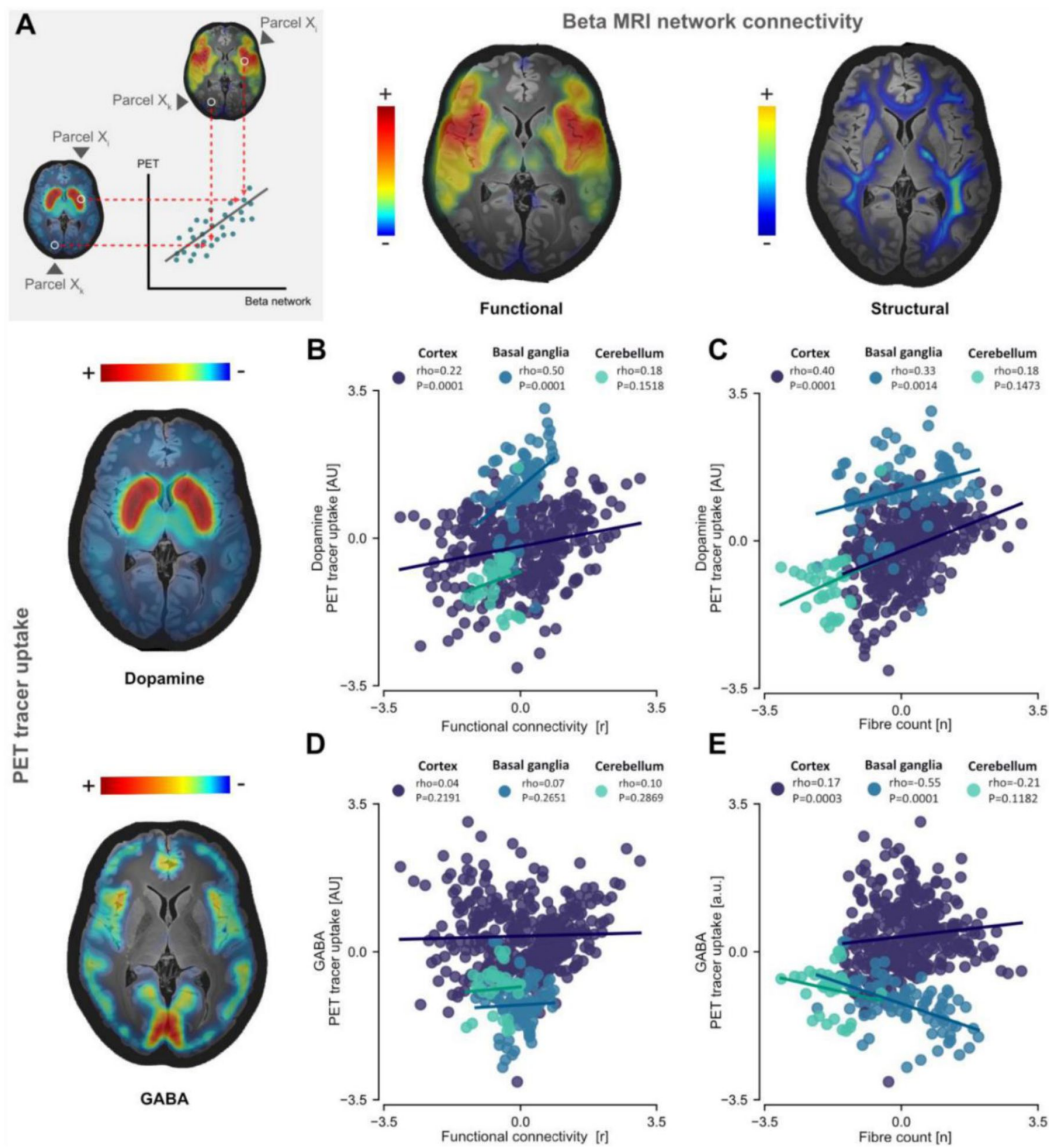


Figure 3

The connectomic beta network correlates with molecular markers of dopamine signalling.

Dopamine uptake significantly correlates with beta functional connectivity (**A**) A methods schematic explaining how correlations were calculated in **Figure 3**. Each point on the scatterplot represents a parcellation from the compound brain atlas. For every parcel X_i , a correlation is calculated for that parcel in the PET scan and in the beta network (**B**) in cortex ($\rho=0.22$, $P=0.0001$) and basal ganglia ($\rho=0.50$, $P=0.0001$) but not cerebellum ($P>0.05$), and beta structural connectivity (**C**) in cortex ($\rho=0.40$, $P=0.0001$) and basal ganglia ($\rho=0.33$, $P=0.0014$) but not cerebellum ($P>0.05$). GABA was used as a control molecule which revealed no significant correlations with beta functional connectivity (**D**) in cortex, basal ganglia or cerebellum ($P>0.05$). Beta structural connectivity (**E**) revealed weaker positive correlation with GABA in cortex ($\rho=0.17$, $P=0.0003$), a negative correlation in the basal ganglia ($\rho=-0.55$, $P=0.0001$) and no correlation in cerebellum ($P>0.05$).

cortex³⁵. Our connectomic characterisation of the beta network provides a network basis for these functions and highlights that similar synchronisation phenomena occur in brain areas associated with vastly different functions, such as prefrontal, cingulate and parietal areas, which give rise to parallel striatal and other subcortical projections. In line with this more complex picture, direct measurement of dopamine concentration in non-human primates revealed specific interactions between dopamine release, beta oscillations, reward value and motor control, depending on contextual information and striatal domain³⁶. This shows that the relationship of dopamine and beta activity is not solely associated with either reward or movement and depends on where in the striatum beta activity is recorded. Thus, an obvious differentiator of these reported functions can be the specific brain region in which beta activity has been recorded from. However, the aim of our study was the opposite, namely to provide evidence for a shared network that unifies these functions.

An authoritative perspective article has proposed that the unifying function of beta oscillations is to signal the maintenance of “status quo”, e.g. to indicate whether or not a motor or cognitive state should be changed or not³⁷. We hypothesise that, this “status quo” hypothesis could be equally or maybe even more adequately posed on the neural level. Namely, it could provide insights to what degree a certain activity pattern or synaptic connection is to be strengthened or weakened, in light of neural learning. We propose that this putative function can be contextualised in a recently described framework of neural reinforcement, that serves to orchestrate the re-entrance and refinement of neural population dynamics for the production of neural trajectories³⁸. On the basis of our observed network architecture described for beta oscillations in this article, we speculate that the “status quo” hypothesis and each of the abovementioned behavioural correlates can be summarized to reflect states with either loss or invigoration of neural reinforcement³⁹, the brain’s intrinsic capacity to modulate the stability of population dynamics via cortico-striatal projections that are orchestrated by dopamine. To simplify this, maintenance of a behavioural, cognitive or emotional “status quo” may be reflected by beta activity on the neural circuit level, which in turn relates to the stability of neural dynamics in frontostriatal circuits that can be dynamically modulated by tonic and phasic changes of dopamine release. If beta is high, the neural dynamics remain stable. If beta is low, new neural trajectories may arise that lead to neural and behavioural state changes and neural learning. Thus, we speculate that the observed beta network and its relation to molecular markers of dopamine in this study may indicate that beta modulation occurs in situations that are associated with neural learning and optimisation of neural trajectories, or lack thereof. Notably, many functional correlates of beta activity could be summarised under the “status quo” hypothesis, but some functions are not easily interpreted in this framework, such as the specific observations in the context of memory processing²⁸, or regulation of emotions and depression. On the other hand, the role for dopaminergic reinforcement and its implication in these processes is robustly established, both in memory⁴⁰ and the pathophysiology of depression⁴¹. At this point, this discussion remains speculative, but advances in neuroscientific research methods that combine neurophysiology with neurotransmitter release may soon provide new evidence in favour or in opposition to these concepts.

Implications for neurotechnology and clinical brain computer interfaces

A key component of clinical brain-computer interfaces (BCIs) is the bidirectional communication between brain activity and closed-loop control of neurotherapies. The robust association of the hypodopaminergic state in Parkinson’s disease with excessive beta synchrony has led to the development of a closed-loop neurostimulation therapy using beta activity as a feedback signal to adjust administration of deep brain stimulation (DBS) in real-time to therapeutic demand⁴². This approach, also termed adaptive DBS, was first demonstrated based on cortical beta activity that was used to adapt pallidal DBS in the MPTP non-human primate model of PD⁴³. It was quickly translated to first experimental studies using subcortical beta signals in human patients⁴⁴,

followed by further research using more complex cortical and subcortical sensing setups and biomarker combinations^{45,46}. Practically, beta activity is used in this novel therapeutic approach to infer the dopaminergic tone and consequential therapeutic demand in PD, for rapid adaptation of therapeutic delivery to the individual state that the patients are facing in real-time. Our study provides support for the validity of this approach, by providing further evidence for a relationship of dopamine and beta activity at the global circuit level. More importantly, our findings may suggest that this relationship could be of greater utility of neurotechnology to treat other dopaminergic disorders. In fact, this notion is in part already backed by empirical data, that shows that beta can be a valuable biomarker for depression severity as identified by machine learning algorithms in invasive neurophysiological recordings from brain implants⁷. Given our observation that beta activity is the most widespread cortical brain rhythm, it is foreseeable that future experimental in-human trials aiming to test the utility of invasive neurophysiology for the adaptation of neurotherapies will be capable to track beta dynamics in higher frequency (sampling rate) and higher precision than any other brain rhythm. Our findings could encourage clinical neuroscience and neural engineering teams to conceptualise the dynamic fluctuations of beta oscillations in context of cortico-striatal processing and dopaminergic modulation, which could lead to new hypothesis driven treatment avenues. Specifically, our demonstration that the distributed occurrence of beta activity shares a brain network that is modulated by dopamine, may motivate the use of beta oscillations to control BCIs or closed-loop neurostimulation for emerging indications related to dopaminergic dysfunction.

Limitations

The most important limitation of this project is aggregated and derived nature of the used brain data. Three independent datasets from different study cohorts, were leveraged to fuse the invasive neurophysiology with MRI and PET based neuroimaging. On the one hand, the independence of the source data may be seen as a form of validation, e.g. for functional and structural connectivity that showed highly overlapping findings while stemming from entirely different cohorts. On the other hand, precisely this aspect severely limits the generalisability of our results to the level of the individual human subject. Similarly, the foundation of the results is built on patients undergoing brain surgery for epilepsy, a severe intractable brain disorder. But at the same time, the reported observations follow a long line of research and corroborate anecdotal or clinical knowledge, that beta activity is a ubiquitous phenomenon. In this light, our study provides an unprecedented and methodologically robust quantitative approach, by using only localised bipolar invasive recordings with oscillatory activity parametrised for optimal comparability. We believe that this rigorous neurophysiological methodology, alongside the fact that the brain signals have been meticulously controlled for pathological activity, by several experienced neurophysiologists, justifies its use and superiority over aggregates of non-invasive recordings from healthy subjects. Another limitation in our study is the fact that the statistical approach for the comparison of beta and alpha networks and even for multiple peaks within the same frequency band follows a winner takes all logic that is, by definition, a simplification, as most areas will contribute to more than one spatio-spectrally distinct oscillatory network. Specifically, while multiple peaks within or across frequency bands could be present in each channel, we decided to allocate this channel to only the frequency band containing the highest peak amplitude. Alternatively, channels could have been allocated to multiple networks, contrasting channels that display vs. do not display peaks in a certain range.

Alternative approaches could yield different results, e.g. reusing channels for each peak that is observable and contrasting them to channels where such peak was not present. However, in our study the majority of channels exhibited beta activity, even if peaks were of low amplitude, which we believe would have led to less interpretable results. Nevertheless, a previous study combining invasive basal ganglia neurophysiology with non-invasive cortical magnetoencephalography (MEG) recordings has identified the strongest oscillatory network of the internal pallidum, to connect to frontal cortex in the beta frequency range, which is in line with our findings⁴⁷. However, additional networks, such as an alpha network with the cerebellum and a theta network

with the temporal lobe were also observable in these parallel recordings. Thus, the presented MRI based statistical maps in this study can be interpreted to reflect shared large-scale and robustly estimable whole-brain circuits of dominant beta vs. alpha oscillations but should not be misunderstood to reflect mutual exclusivity for the presence or communication in additional brain rhythms. In fact, when looking at individual parcels of structural connectivity, it is noticeable, that both beta and alpha networks share strong connections to cingulate cortex. This is interesting, because previous invasive neurophysiology studies in patients undergoing deep brain stimulation surgery for depression have reported both excessive alpha and beta activity as potential biomarkers for depressive symptoms in the human cingulum³⁵, potentially related to differential modulation of neurotransmitters involved in the disease. One impactful study reported that beta oscillatory sub-networks of Amygdala and hippocampus could reflect human variations in mood³⁴. This is interesting, but highlights another relevant limitation of our study, namely that recordings in different areas were stemming from different patients and thus, such sub-network analyses on the oscillatory level could not be conducted.

Another limitation arises as a recurring controversy in fMRI research (cite) is the issue insufficient knowledge about negative correlations and their exact scientific implications. At present, there are no straightforward answers about what such correlations actually mean; and therefore we have been hesitant in drawing conclusions relating to the signs of fMRI connectivity, and instead, treat it as a spatial fingerprint.

It should also be noted that in this project, the data are static, but beta oscillations and dopamine are dynamic⁴⁸. Therefore, while we cannot infer the direction of the relationship, it is very likely that the relationship would be negative implying higher beta would mean lower dopamine. Finally, our study is limited to resting state activity and thus does not allow for interpretation of functional changes with relation to behaviour or neural processes, that may individually be associated with other or additional brain rhythms. Most notably, gamma band oscillations are known to be reflective of local processing but notoriously absent in resting state activity and thus are likely equally or even more ubiquitous than beta activity. However, we do not see this as a contradiction, as resting beta activity provides insights into a different state of the cortical brain area, potentially more associated with global circuit synchrony, while gamma band activity may be more reflective of local activity in the cortical microcircuit. The aim of this study was to elucidate the circuit architecture of beta oscillations, which is why insights from this study for other frequency bands are limited. Future research investigating the specific circuits of theta, alpha and gamma oscillations and their relationship with neurotransmitter uptake could yield new important insights on the networks underlying human brain rhythms.

Conclusion

In summary, our study suggests that the beta rhythm is the most widespread cortical oscillatory activity pattern in the resting brain, that shares connectivity with the basal ganglia via a whole brain network that is modulated by dopamine. We interpret our findings to provide a unifying circuit-based integration of previous reports on the functional role of beta oscillations. In the future, our findings may inspire and encourage the investigation of beta oscillations for the use as a feedback signal for neurotechnological interventions in the treatment of dopaminergic disorders.

Methods

Atlas of invasive electrophysiology

The collection of data, cohort and electrophysiology of the MNI Open iEEG Atlas (<https://mni-open-ieegatlas.research.mcgill.ca/>) is as described in the work of Frauscher and colleagues⁹. Intracranial EEG (iEEG) recordings from the healthy hemispheres of epilepsy patients were

collected from three hospitals across Canada. Inclusion criteria for the recordings included that it must 1) contain at least one channel with normal activity, 2) have peri-implantation imaging, 3) be recorded after a minimum of 72 hours post-implantation for stereo-EEG electrodes or 1 week for subdural grids or strips, and 4) have a minimum sampling frequency of 200 Hz. The recordings were analysed to identify frequencies in the classical Berger frequency bands (0.3–70 Hz). Tissue containing channels exhibiting normal activity was classified as normal or healthy tissue. This meant it was outside of the zone of seizure onset, i.e., no interictal epileptic discharge and no discharges or slow wave anomalies were found there. Ethical approval was granted at the MNI (Montreal Neurological Institute) as lead ethics organization (REB vote: MUHC-15-950). Electrode locations were registered to a common stereotaxic space.

Oscillatory analysis and neurophysiological pre-processing

Data from the iEEG atlas were preprocessed using the Statistical Parametric Mapping toolbox (<https://www.fil.ion.ucl.ac.uk/spm12/>). The original data contained zero padded intervals between signal fragments which were deleted. Next the data were Fourier transformed to obtain a time-frequency signal using Morlet Wavelets with seven cycles. All time-frequency signals were averaged over time to obtain one frequency spectrum of 60 second sections of iEEG measurements. The resulting power spectra were quantified in terms of spectral peak properties and background level activity using FOOOF¹⁰ with the following parameters; limits for possible peak widths were [0.5, 12 Hz], the maximum number of detectable peaks were set to infinite so as to not limit the number of peaks detected, the minimum peak height was 0 while the peak threshold was within 2 standard deviations of modelled power. Peak height was extracted using the pw parameter, which depicts peak amplitude after subtraction of any aperiodic activity. In case of multiple peaks within the same region, we used only the highest peak amplitude. Extracted were z-scored within electrodes, within subjects and within the whole dataset to quantify neural activity from several frequency bands of the spectrum. Frequency-specific activity was quantified using maximum peak amplitude for specific frequency ranges. The maximum peak amplitude was extracted from the z-scored parameters for four frequency bands; theta (4–8 Hz), alpha (8–12 Hz), beta (13–35 Hz), and gamma (30–100 Hz).

Connectivity analysis

Resting state functional connectivity data was collected from 1,000 healthy individuals using a 3T Siemens MRI scanner as part of the Brain Genomics Superstruct Project. Preprocessing included global signal regression and spatial smoothing at 6mm full width at half maximum. The group connectome for structural connectivity was obtained from multishell diffusion-weighted and T2-weighted imaging data subjects from the Human Connectome Project⁴⁹. The imaging data were acquired using a custom MRI scanner with enhanced gradients. Whole brain tractography fibre sets were generated using a generalised q-sampling imaging algorithm implemented in DSI Studio. Connectivity analyses were performed in MATLAB using normative connectomes calculated with Lead-Mapper⁵⁰. Three-dimensional spherical nifti files representing channel locations were extracted and used as seeds to calculate connectivity maps.

The fMRI and dMRI data underwent processing using the Lead Connectome Mapper. Every contact location from the iEEG atlas was represented by a spherical seed nifti image (r=5mm). To minimise the impact of noise on the results, all maps underwent smoothing with a Gaussian kernel (standard deviation = 8mm). Subsequently, the fMRI maps were masked with a grey matter mask to avoid the inclusion of white matter correlations, which would have introduced artifacts.

A functional connectivity map, i.e., a mapping of the functional connectivity (based on a normative connectome) between a seed region and every other brain region, was derived for each individual channel (N=1772). To do so, each channel was used as a seed region and its respective functional connectivity map was calculated, resulting in 1772 connectivity maps, one for every channel. These maps were subdivided based on the channel's dominant frequency band. This resulted in

277 functional connectivity maps for the theta, 397 for the alpha and 1005 for the beta band. Group level statistics were performed for connectivity maps of each frequency band. All functional connectivity maps were averaged in SPM ImCalc to reveal a group-level functional network for each frequency band.

To investigate differences between beta dominant and alpha dominant functional connectivity networks, a two sample t-test was calculated for the condition where beta was greater than alpha and vice versa using SPM. Here, the connectivity maps from each dominant channel (1005 beta functional connectivity maps and 397 alpha connectivity maps) Estimation of model parameters yielded t-values for each voxel, indicating the strength and direction of differences between the two contrasts (beta > alpha, alpha > beta). To address the issue of multiple comparisons, we applied Family-Wise Error (FWE) correction, adjusting significance thresholds such that only voxels with $p < 0.05$ would be included.

Like functional connectivity maps, structural connectivity maps (N=1772) were calculated for each individual channel. These were again subdivided based on dominant frequency band per channel and gave a resulting set of structural connectivity maps for each frequency band of 277 (theta), 397 (alpha) and 1005 (beta). Each map contained a set of fibre tracts that were identified as correlated with the specific channel through the structural connectivity analysis. Therefore, to investigate group level statistics, the maps had to be summed and not averaged as in the case of functional connectivity. The resulting structural networks for each frequency band were hence a sum calculated on SPM ImCalc of all identified fibres for all channels for a given frequency band.

Neurotransmitter uptake

To investigate dopamine uptake, an open dataset of neurotransmitter uptake was used an open data set of PET data measuring neurotransmitter uptake¹⁸. Exclusively healthy samples of subjects were selected for various dopamine receptors such as D1⁵¹ (healthy controls (HC)=13), D2^{52,53} (HC=34), the dopamine transporter (DAT)^{54,55} (HC=180) and fluorodopa (FDOPA)⁵⁶ (HC=12). To study the broad effects of dopamine, all dopamine receptor PET scans were resliced and averaged using the SPM ImCalc function. A spatial normalisation was then applied to finally create a unified aggregate image of all available dopamine receptors. A custom master parcellation in MNI space was created in Matlab using SPM functions by combining three existing parcellations to include cortical regions¹⁹, structures of the basal ganglia²⁰ and cerebellar regions²¹. Regions that were (partially) overlapping between the atlases were only selected once. The final compound parcellation had 476 regions in total. This parcellation was applied to both PET and unthresholded structural and functional connectivity maps using SPM and custom code. This allowed for the calculation of spatial correlations, providing a statistical measure of spatial similarity of the PET intensity and MRI connectivity distributions. For this, Spearman's ranked correlations were used to calculate correlations between the PET images, such as the dopamine aggregate map and both functional and structural beta connectivity networks (**Figure 3**). The analysis was repeated for individual tracers showing similar results (Supplementary figure 5). Finally, to validate these results, a control analysis was performed using a GABA PET scan from the same open dataset of neurotransmitter uptake following the same pipeline (**figure 2A, 2B**).

Data availability

All data used in the course of this work were publicly available open datasets and can be found online.

1. The iEEG atlas is available on the McGill database and can be accessed via their publication (https://www.mcgill.ca/frauscher-lab/files/frauscher-lab/frauscher_brain2018.pdf).
2. The normative connectomes are available within the Lead-DBS software (www.lead-dbs.org).

3. The PET scans for dopamine uptake as well as their data pipeline are available on the Network Neuroscience Lab GitHub repository(https://github.com/netneurolab/hansen_receptors .

Code availability

All original code used to preprocess and analyze these data, and create the main and supplementary results is openly available on a GitHub repository (https://github.com/meera-cm/iEEG_connectomics .

Additional code from the open datasets is available:

1. Lead-DBS and Lead-Connectome Mapper: (<https://github.com/leaddbs/leaddbs> )
2. Hansen receptors: (https://github.com/netneurolab/hansen_receptors/tree/main/code )

Acknowledgements

Alessia Cavallo, Dr. Till Dembek, Dr. Konstantin Butenko and Dr. Garance Meyer for their support in the preparation of the manuscript.

Additional information

Author contributions

WJN and MC formulated and conceptualised the experiments. Methods and data analysis were designed by WJN and MC and executed by MC, LW, JV and TM. Visualisations were done by MC, NR, RK and WJN. The project was supervised by WJN. The manuscript was written by MC and WJN.

Funding statement

The study was funded through the Hertie Network of Excellence in Clinical Neuroscience fellowship to WJN. WJN further received funding from the European Union (ERC, ReinforceBG, project 101077060), Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project-ID 424778381 – TRR 295 and the Bundesministerium für Bildung und Forschung (BMBF, project FKZ01GQ1802).

Declaration of interests

WJN received honoraria for consulting from InBrain – Neuroelectronics that is a neurotechnology company and honoraria for talks from Medtronic that is a manufacturer of deep brain stimulation devices unrelated to this manuscript.

Note

This reviewed preprint has been updated to correct an author name.

References

1. Fries P (2015) **Rhythms for Cognition: Communication through Coherence** *Neuron* **88**:220–235
2. Hammond C., Bergman H., Brown P (2007) **Pathological synchronization in Parkinson's disease: networks, models and treatments** *Trends Neurosci* **30**:357–364
3. Paulo D. L., et al. (2023) **Corticostriatal beta oscillation changes associated with cognitive function in Parkinson's disease** *Brain* **146**:3662–3675
4. Iskhakova L., et al. (2021) **Modulation of dopamine tone induces frequency shifts in cortico-basal ganglia beta oscillations** *Nat Commun* **12**
5. Berke J. D (2018) **What does dopamine mean?** *Nat Neurosci* **21**:787–793
6. Hanslmayr S., Staresina B. P., Bowman H (2016) **Oscillations and Episodic Memory: Addressing the Synchronization/Desynchronization Conundrum** *Trends in Neurosciences* **39**:16–25
7. Alagapan S., et al. (2023) **Cingulate dynamics track depression recovery with deep brain stimulation** *Nature* **622**:130–138
8. Hedrich T., Pellegrino G., Kobayashi E., Lina J. M., Grova C (2017) **Comparison of the spatial resolution of source imaging techniques in high-density EEG and MEG** *NeuroImage* **157**:531–544
9. Frauscher B., et al. (2018) **Atlas of the normal intracranial electroencephalogram: neurophysiological awake activity in different cortical areas** *Brain* **141**:1130–1144
10. Donoghue T., et al. (2020) **Parameterizing neural power spectra into periodic and aperiodic components** *Nat Neurosci* **23**:1655–1665
11. Rolls E. T., Huang C.-C., Lin C.-P., Feng J., Joliot M (2020) **Automated anatomical labelling atlas 3** *NeuroImage* **206**
12. Pellegrini F., Delorme A., Nikulin V., Haufe S (2023) **Identifying good practices for detecting inter-regional linear functional connectivity from EEG** *NeuroImage* **277**
13. Engel A. K., Moll C. K. E., Fried I., Ojemann G. A (2005) **Invasive recordings from the human brain: clinical insights and beyond** *Nat Rev Neurosci* **6**:35–47
14. Fox M. D (2018) **Mapping Symptoms to Brain Networks with the Human Connectome** *New England Journal of Medicine* **379**:2237–2245
15. Lofredi R., et al. (2021) **Subthalamic stimulation impairs stopping of ongoing movements** *Brain* **144**:44–52
16. Neudorfer C., et al. (2023) **Lead-DBS v3.0: Mapping deep brain stimulation effects to local anatomy and global networks** *NeuroImage* **268**

17. Lofredi R., et al. (2023) **Subthalamic beta bursts correlate with dopamine-dependent motor symptoms in 106 Parkinson's patients** *npj Parkinsons Dis* **9**:1–9
18. Hansen J. Y., et al. (2022) **Mapping neurotransmitter systems to the structural and functional organization of the human neocortex** *Nat Neurosci* **25**:1569–1581
19. Huang C.-C., Rolls E. T., Feng J., Lin C.-P (2022) **An extended Human Connectome Project multimodal parcellation atlas of the human cortex and subcortical areas** *Brain Struct Funct* **227**:763–778
20. He X., et al. (2020) **Disrupted basal ganglia–thalamocortical loops in focal to bilateral tonic-clonic seizures** *Brain* **143**:175–190
21. Diedrichsen J., Balsters J. H., Flavell J., Cussans E., Ramnani N (2009) **A probabilistic MR atlas of the human cerebellum** *Neuroimage* **46**:39–46
22. Neumann W.-J., Horn A., Kühn A. A (2023) **Insights and opportunities for deep brain stimulation as a brain circuit intervention** *Trends in Neurosciences* **46**:472–487
23. Deffains M., Bergman H (2019) **Parkinsonism-related β oscillations in the primate basal ganglia networks – Recent advances and clinical implications** *Parkinsonism & Related Disorders* **59**:2–8
24. Sharott A., et al. (2005) **Dopamine depletion increases the power and coherence of β -oscillations in the cerebral cortex and subthalamic nucleus of the awake rat** *European Journal of Neuroscience* **21**:1413–1422
25. Romo R., Schultz W (1990) **Dopamine neurons of the monkey midbrain: contingencies of responses to active touch during self-initiated arm movements** *Journal of Neurophysiology* **63**:592–606
26. Patriarchi T., et al. (2018) **Ultrafast neuronal imaging of dopamine dynamics with designed genetically encoded sensors** *Science* **360**
27. Coddington L. T., Dudman J. T (2019) **Learning from Action: Reconsidering Movement Signaling in Midbrain Dopamine Neuron Activity** *Neuron* **104**:63–77
28. Miller E. K., Lundqvist M., Bastos A. M (2018) **Working Memory 2.0** *Neuron* **100**:463–475
29. Leventhal D. K., et al. (2012) **Basal Ganglia Beta Oscillations Accompany Cue Utilization** *Neuron* **73**:523–536
30. Meyer L (2018) **The neural oscillations of speech processing and language comprehension: state of the art and emerging mechanisms** *European Journal of Neuroscience* **48**:2609–2621
31. Pierrieau E., Berret B., Lepage J.-F., Bernier P.-M (2023) **From Motivation to Action: Action Cost Better Predicts Changes in Premovement Beta-Band Activity than Speed** *J. Neurosci* **43**:5264–5275
32. Yin Z., et al. (2023) **Pathological pallidal beta activity in Parkinson's disease is sustained during sleep and associated with sleep disturbance** *Nat Commun* **14**

33. Jabbi M., et al. (2015) **Convergent BOLD and Beta-Band Activity in Superior Temporal Sulcus and Frontolimbic Circuitry Underpins Human Emotion Cognition** *Cerebral Cortex* **25**:1878–1888
34. Kirkby L. A., et al. (2018) **An Amygdala-Hippocampus Subnetwork that Encodes Variation in Human Mood** *Cell* **175**:1688–1700
35. Clark D. L., Brown E. C., Ramasubbu R., Kiss Z. H. T (2016) **Intrinsic Local Beta Oscillations in the Subgenual Cingulate Relate to Depressive Symptoms in Treatment-Resistant Depression** *Biological Psychiatry* **80**:e93–e94
36. Schwerdt H. N., et al. (2020) **Dopamine and beta-band oscillations differentially link to striatal value and motor control** *Sci Adv* **6**
37. Engel A. K., Fries P (2010) **Beta-band oscillations—signalling the status quo?** *Current Opinion in Neurobiology* **20**:156–165
38. Athalye V. R., Carmenta J. M., Costa R. M (2020) **Neural reinforcement: re-entering and refining neural dynamics leading to desirable outcomes** *Curr Opin Neurobiol* **60**:145–154
39. Cavallo A., Wolf-Julian N (2023) **Dopaminergic Reinforcement in the Motor System: Implications for Parkinson’s Disease and Deep Brain Stimulation** *Authorea* <https://doi.org/10.22541/au.169536617.79800198/v1>
40. Rossato J. I., Bevilacqua L. R. M., Izquierdo I., Medina J. H., Cammarota M (2009) **Dopamine Controls Persistence of Long-Term Memory Storage** *Science* **325**:1017–1020
41. Dunlop B. W., Nemeroff C. B (2007) **The Role of Dopamine in the Pathophysiology of Depression** *Archives of General Psychiatry* **64**:327–337
42. Neumann W.-J., Gilron R., Little S., Tinkhauser G (2023) **Adaptive Deep Brain Stimulation: From Experimental Evidence Toward Practical Implementation** *Movement Disorders* **38**:937–948
43. Rosin B., et al. (2011) **Closed-Loop Deep Brain Stimulation Is Superior in Ameliorating Parkinsonism** *Neuron* **72**:370–384
44. Little S., et al. (2013) **Adaptive deep brain stimulation in advanced Parkinson disease** *Annals of Neurology* **74**:449–457
45. Gilron R., et al. (2021) **Long-term wireless streaming of neural recordings for circuit discovery and adaptive stimulation in individuals with Parkinson’s disease** *Nat Biotechnol* **39**:1078–1085
46. Oehr C. R., et al. (2023) **Personalized chronic adaptive deep brain stimulation outperforms conventional stimulation in Parkinson’s disease** *medRxiv* <https://doi.org/10.1101/2023.08.03.23293450>
47. Neumann W.-J., et al. (2015) **Cortico-pallidal oscillatory connectivity in patients with dystonia** *Brain* **138**:1894–1906
48. Tinkhauser G., et al. (2017) **The modulatory effect of adaptive deep brain stimulation on beta bursts in Parkinson’s disease** *Brain* **140**:1053–1067

49. Setsompop K., et al. (2013) **Pushing the limits of in vivo diffusion MRI for the Human Connectome Project** *NeuroImage* **80**:220–233
50. Horn A., Kühn A. A (2015) **Lead-DBS: A toolbox for deep brain stimulation electrode localizations and visualizations** *NeuroImage* **107**:127–135
51. Kaller S., et al. (2017) **Test–retest measurements of dopamine D1-type receptors using simultaneous PET/MRI imaging** *Eur J Nucl Med Mol Imaging* **44**:1025–1032
52. Alakurtti K., et al. (2015) **Long-Term Test–Retest Reliability of Striatal and Extrastriatal Dopamine D2/3 Receptor Binding: Study with [11C]Raclopride and High-Resolution PET** *J Cereb Blood Flow Metab* **35**:1199–1205
53. Smith C. T., et al. (2019) **Partial-volume correction increases estimated dopamine D2-like receptor binding potential and reduces adult age differences** *J Cereb Blood Flow Metab* **39**:822–833
54. Dukart J., et al. (2018) **Cerebral blood flow predicts differential neurotransmitter activity** *Sci Rep* **8**
55. Sasaki T., et al. (2012) **Quantification of Dopamine Transporter in Human Brain Using PET with 18F-FE-PE2I** *Journal of Nuclear Medicine* **53**:1065–1073
56. Gómez F. J. G., Huertas I., Ramírez J. A. L., Solís D. G (2018) **Elaboración de una plantilla de SPM para la normalización de imágenes de PET con 18F-DOPA** *Imagen Diagnóstica* **9**:23–25

Editors

Reviewing Editor

Jonas Obleser

University of Lübeck, Lübeck, Germany

Senior Editor

Michael Frank

Brown University, Providence, United States of America

Reviewer #1 (Public review):

The study by Chikermane and colleagues investigates functional, structural, and dopaminergic network substrate of cortical beta oscillations (13-30 Hz). The major strength of the work lies in the methodology taken by the authors, namely a multimodal lesion network mapping. First, using invasive electrophysiological recordings from healthy cortical territories of epileptic patients they identify regions with highest beta power. Next, they leverage open access MRI data and PET atlases and use the identified high-beta regions as seeds to find (1) the whole-brain functional and structural maps of regions that form the putative underlying network of high-beta regions and (2) the spatial distribution of dopaminergic receptors that show correlation with nodal connectivity of the identified networks. These steps are achieved by generating aggregate functional, structural, and dopaminergic network maps using lead-DBS toolbox, and by contrasting the results with those obtained from high-alpha regions. The main findings are:

- (1) Beta power is strongest across frontal, cingulate, and insular regions in invasive electrophysiological data, and these regions map onto a shared functional and structural network.
- (2) The shared functional and structural networks show significant positive correlations with dopamine receptors across cortex and basal ganglia (which is not the case for alpha, where correlations are found with GABA).

<https://doi.org/10.7554/eLife.97184.2.sa3>

Reviewer #2 (Public review):

Summary:

This is a very interesting paper that leveraged several publicly available datasets: invasive cortical recording in epilepsy patients, functional and structural connectomic data, and PET data related to dopaminergic and gaba-ergic synapses. These were combined to create a unified hypothesis of beta band oscillatory activity in the human brain. They show that beta frequency activity is ubiquitous, and does not just occur in sensorimotor areas. Cortical regions where beta oscillations predominated had high connectivity to regions that are high in dopamine re-update.

Strengths:

The authors leverage and integrate three publicly available human brain datasets in a creative way. These public datasets are powerful tools for human neuroscience, and it is innovative to combine these three types of data into a common brain space to generate novel findings and hypotheses. Findings are nicely controlled by separately examining cortical regions where alpha predominates (which have a different connectivity pattern). GABA uptake from PET studies is used as a control for the specificity of the relationship between beta activity and dopamine uptake. There is much interest in synchronized oscillatory activity as a mechanism of brain function and dysfunction, but the field is short on unifying hypotheses of why particular rhythms predominate in particular regions. This paper contributes nicely to that gap. It is ambitious in generating hypotheses, particularly that modulation of beta activity may be used as a "proxy" for modulating phasic dopamine release.

Weaknesses:

As the authors point out, the use of normative data is excellent for exploring hypotheses but does not address or explore individual variations which could lead to other insights. It is also biased to resting state activity; maps of task related activity (if they were available) might show different findings.

Challenges:

In the Discussion, the authors do a fairly deep dive into the implications of their findings, particularly with respect to the hypothesis that beta band activity "preserves the status quo", and with respect to the use of beta band activity in controlling brain-machine interfaces. Mechanistically and theoretically oriented readers might gain rewarding new insights by a careful read of the discussion, but full appreciation of their deep dive may require real time interaction with the authors.

<https://doi.org/10.7554/eLife.97184.2.sa2>

Reviewer #3 (Public review):**Summary:**

In this paper, Chikermane et al. leverage a large open dataset of intracranial recordings (sEEG or ECoG) to analyze resting state (eyes closed) oscillatory activity from a variety of human brain areas. The authors identify a dominant proportion of channels in which beta band activity (12-30Hz) is most prominent, and subsequently seek to relate this to anatomical connectivity data by using the sEEG/ECoG electrodes as seeds in a large set of MRI data from the human connectome project. This reveals separate regions and white matter tracts for alpha (primarily occipital) and beta (prefrontal cortex and basal ganglia) oscillations. Finally, using a third available dataset of PET imaging, the authors relate the parcellated signals to dopamine signaling as estimated by spatial uptake patterns of dopamine, and reveal a significant correlation between the functional connectivity maps and the dopamine reuptake maps, suggesting a functional relationship between the two.

Strengths:

Overall, I found the paper well justified, focused on an important topic and interesting. The authors' use of 3 different open datasets was creative and informative, and it significantly adds to our understanding of different oscillatory networks in the human brain, and their more elusive relation with neuromodulator signaling networks by adding to our knowledge of the association between beta oscillations and dopamine signaling. Even my main comments about the lack of a theta network analysis and discussion points are relatively minor, and I believe this paper is valuable and informative.

Weaknesses:

The analyses were adequate, and the authors cleverly leverage these different datasets to build an interesting story. The main aspect I found missing (in addition to some discussion items, see below) was an examination of the theta network. Theta oscillations have been involved in a number of cognitive processes including spatial navigation and memory, and have been proposed to have different potential originating brain regions, and it would be informative to see how their anatomical networks (e.g. as in Fig. 2) look like under the author's analyses.

The authors devote a significant portion of the discussion to relating their findings to a popular hypothesis for the function of beta oscillations, the maintenance of the "status quo", mostly in the context of motor control. As the authors acknowledge, given the static nature of the data and lack of behavior, this interpretation remains largely speculative and I found it a bit too far-reaching given the data shown in the paper. In contrast, I missed a more detailed discussion on the growing literature indicating a role for beta in mood (e.g. in Kirkby et al. 2018), especially given the apparent lack of hippocampal and amygdala involvement in the paper, which was surprising.

<https://doi.org/10.7554/eLife.97184.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

The study by Chikermane and colleagues investigates the functional, structural, and dopaminergic network substrates of cortical beta oscillations (13-30 Hz). The major

strength of the work lies in the methodology taken by the authors, namely a multimodal lesion network mapping. First, using invasive electrophysiological recordings from healthy cortical territories of epileptic patients they identify regions with the highest beta power. Next, they leverage open-access MRI data and PET atlases and use the identified high-beta regions as seeds to find (1) the whole-brain functional and structural maps of regions that form the putative underlying network of high-beta regions and (2) the spatial distribution of dopaminergic receptors that show correlation with nodal connectivity of the identified networks. These steps are achieved by generating aggregate functional, structural, and dopaminergic network maps using lead-DBS toolbox, and by contrasting the results with those obtained from high-alpha regions.

The main findings are:

(1) Beta power is strongest across frontal, cingulate, and insular regions in invasive electrophysiological data, and these regions map onto a shared functional and structural network. (2) The shared functional and structural networks show significant positive correlations with dopamine receptors across the cortex and basal ganglia (which is not the case for alpha, where correlations are found with GABA).

Nevertheless, a few clarifications regarding the choice of high-power electrodes and distributions of functional connectivity maps (i.e., strength and sign across cortex and sub-cortex) can help with understanding the results.

We thank the reviewer for this critical expert assessment.

Reviewer #1 (Recommendations For The Authors):

To potentially enhance the quality of the manuscript in the current version, I kindly ask the authors to address the following points:

Major:

(A) Power analysis of electrophysiological data

(1) How were significant peaks identified exactly? I understand that the authors used FOOOF methodology to estimate periodic components of brain activity.

Thank you for pointing us to this lack of clarity. The application of FOOOF consists of the fitting of a one-over-f curve that delineates the aperiodic component followed by the definition of gaussians to fit periodic activity. This allows for extraction of periodic peak power estimates that are corrected for offset and exponent of the one-over-f or non-oscillatory aperiodic component in the spectrum (further information can be found here https://fooof-tools.github.io/fooof/auto_tutorials/plot_02-FOOOF.html). We included all peaks that could be fitted using the process.

How about aperiodic components (Figure 1, PSD plots)?

We share the interest in aperiodic activity with the reviewer. However, given that the primary aim of this study was the description of beta oscillations and the methodology and results presentation is already very complex, we did not include the analysis of aperiodic activity in this manuscript. This could be done in the future and it would surely be interesting to visualize the whole brain connectomic fingerprints of aperiodic exponent and offset. With regard to the purely anatomical description of nonoscillatory aperiodic activity we would like to refer to Figure 8 in Frauscher et al. Brain 2018 (<https://doi.org/10.1093/brain/awy035>) where this is described. We have decided not to include additional information on this matter, because a) we felt that this would further convolute the results and discussion

without directly addressing any of the hypotheses and aims that we set out to tackle and b) the interpretation of aperiodic activity is still a matter of intense research with conflicting results, which warrants very careful considerations of many aspects that again would go beyond the scope of this paper.

In addition, to what degree would the results change if one identified the peaks relative to sites with no peak, similar to Frauscher et al.

Beta activity, the oscillation of interest in our analysis is ubiquitous in the brain. In fact, of 1772 channels, only 21 channels did not exhibit a beta peak detectable with FOOOF. Thus, a comparison of 1751 against 21 would not yield meaningful results. We have therefore decided to focus on the channels in which beta activity is the strongest and dominant observable oscillation.

If the FOOOF approach has some advantages, these should be pointed out or discussed.

FOOOF indeed has the advantage that it provides an objective and reproducible estimation of peak oscillatory activity that accounts for differences in aperiodic activity. To the best of our knowledge, there is no other approach that is nearly as well documented, validated and computationally reproducible.

Changes in manuscript: We have now further clarified the definition of peak amplitudes in the results and methods section and have discussed the use of alternative measures in the limitations section of our manuscript.

Results: “The frequency band with the highest peak amplitude was identified using the extracted peak parameter (pw) for each channel and depicted as the dominant rhythm for the respective localisation (Figure 1).”

Methods: “Peak height was extracted using the pw parameter, which depicts peak amplitude after subtraction of any aperiodic activity.”

Discussion: “Alternative approaches could yield different results, e.g. reusing channels for each peak that is observable and contrasting them to channels where such peak was not present. However, in our study the majority of channels exhibited beta activity, even if peaks were of low amplitude, which we believe would have led to less interpretable results.”

(2) How exactly do the authors deal with channels with more than one peak? Some elaboration on this and how this could potentially impact the results would be appreciated. Sorry if I have missed it.

Indeed, a description of this was lacking so we are very thankful that the reviewer pointed this out. The maximum peak amplitude method was a winner-takes-all approach where in the case of multiple peaks, the peak with the higher amplitude was chosen. This method of course has drawbacks in the form of lost or disregarded peaks and remains a limitation to this study.

Changes in manuscript: We have now clarified this in the methods and results sections, which now read:

Methods: “In case of multiple peaks within the same region, we used only the highest peak amplitude.”

Results: “In case of multiple peaks within the same frequency band, we focused the analysis on the peak with the highest amplitude.”

And added the following to the Limitations section of the discussion:

“Another limitation in our study is the fact that the statistical approach for the comparison of beta and alpha networks and even for multiple peaks within the same frequency band follows a winner takes all logic that is, by definition, a simplification, as most areas will contribute to more than one spatio-spectrally distinct oscillatory network. Specifically, while multiple peaks within or across frequency bands could be present in each channel, we decided to allocate this channel to only the frequency band containing the highest peak amplitude.”

(B) Network mapping

(1) Knowing that fMRI data are preprocessed by regressing the global signal, there are negative correlations across the functional networks. Unfortunately, the distribution, sign, and strength of the correlations are not quantitatively shown in any of the plots. Thus, it is unclear whether, e.g., corticocortical vs. subcortico-cortical correlations differ in strength and/or sign. I think this additional information is important for better understanding the up/down-regulation of beta, e.g., by DA signaling. Some discussion around this point in addition would be insightful, I think.

The referee is touching upon a very important and difficult point, which we have considered very carefully. Global signal regression is a controversial topic and the neurophysiological basis of negative correlations remains to be elucidated. We can justify our use of this approach based on an expert consensus described in Murphy & Fox 2017 (<https://doi.org/10.1016%2Fj.neuroimage.2016.11.052>), which highlights that global signal regression can improve the specificity of positive correlations, improve the correspondence to anatomical connectivity. The truth however is that, we relied on it, because it is the more commonly used and validated approach used in lesion network and DBS connectivity mapping and implemented in the Lead Mapper pipeline. Indeed all connectivity estimates are shown in Supplementary figure 3. We remain hesitant to raise the focus to these points, because of the uncertain underlying neural correlates. However, when looking at the values, it is interesting to note that most key regions of interest exhibit positive connectivity values.

Changes in manuscript: We now point to the supplement containing all connectivity values in the results section more prominently: “All connectivity values including their sign are shown in figures as brain region averages parcellated with the automatic anatomical labelling atlas in supplementary figures 2&3.”

(2) I assume no thresholding is applied to the functional connectivity maps (in a graph-theoretical sense). Please clarify (this is also related to the comment above, in particular, the strength of correlations).

Indeed, we demonstrate SPM maps using family wise error corrected stats in figure 2, but all further analyses were performed on unthresholded maps as correctly pointed out by the referee.

Changes in manuscript:

Results: “Specifically, we analysed to what degree the spatial uptake patterns of dopamine, as measurable with fluorodopa (FDOPA; cohort average of 12 healthy subjects) and other dopamine signalling related tracers that bind D1/D2 receptors (average of N=17/44 respectively healthy subjects) or the dopamine transporter (DAT; cohort average of N=180 healthy subjects) were correlated with the unthresholded MRI connectivity maps.”

Methods: “This parcellation was applied to both PET and unthresholded structural and functional connectivity maps using SPM and custom code.”

Minor

(1) Methods, Connectivity analysis: The description of (mass-univariate) GLM analysis is confusing. The maps underwent preprocessing? Which preprocessing steps are meant here? What is the dependent variable and what are the predictors exactly?

We thank the reviewer for catching this error in our methods. We apologise for the confusion and mistake and thank the reviewer for catching it. Indeed, we have used t-tests without further preprocessing instead of a GLM.

Changes in manuscript: The respective section has been removed from the methods section and intermediate steps have been clarified. The section now reads: "To investigate differences between beta dominant and alpha dominant functional connectivity networks, a two sample t-test was calculated for the condition where beta was greater than alpha and vice versa using SPM. Here, the connectivity maps from each dominant channel (1005 beta functional connectivity maps and 397 alpha connectivity maps) Estimation of model parameters yielded t-values for each voxel, indicating the strength and direction of differences between the two contrasts (beta > alpha, alpha > beta). To address the issue of multiple comparisons, we applied Family-Wise Error (FWE) correction, adjusting significance thresholds such that only voxels with $p < 0.05$ would be included."

(2) I encourage the authors to find a better (visual) way of reporting Table 1, to make the main observations easier to grasp and compare (maybe a two-dimensional bar plot? Or color-coding the cells?)

Reply: Thank you for your suggestion to improve the table, the new table is adjusted to the recommended changes to make it more readable.

Reviewer #2 (Public Review):

Summary:

This is a very interesting paper that leveraged several publicly available datasets: invasive cortical recording in epilepsy patients, functional and structural connectomic data, and PET data related to dopaminergic and gaba-ergic synapses. These were combined to create a unified hypothesis of beta band oscillatory activity in the human brain. They show that beta frequency activity is ubiquitous, not just in sensorimotor areas, and cortical regions where beta predominated had high connectivity to regions high in dopamine re-uptake.

Strengths:

The authors leverage and integrate three publicly available human brain datasets in a creative way. While these public datasets are powerful tools for human neuroscience, it is innovative to combine these three types of data into a common brain space to generate novel findings and hypotheses. Findings are nicely controlled by separately examining cortical regions where alpha predominates (which have a different connectivity pattern). GABA uptake from PET studies is used as a control for the specificity of the relationship between beta activity and dopamine uptake. There is much interest in synchronized oscillatory activity as a mechanism of brain function and dysfunction, but the field is short on unifying hypotheses of why particular rhythms predominate in particular regions. This paper contributes nicely to that gap. It is ambitious in generating hypotheses, particularly that modulation of beta activity may be used as a "proxy" for modulating phasic dopamine release.

Weaknesses:

As the authors point out, the use of normative data is excellent for exploring hypotheses but does not address or explore individual variations which could lead to other insights. It is also biased to resting state activity; maps of task-related activity (if they were available) might show different findings.

The figures, results, introduction, and methods are admirably clear and succinct but the discussion could be both shorter and more convincing.

Reviewer #2 (Recommendations For The Authors):

The tone of the discussion is excessively lofty and abstract, and hard to follow in places. Specific examples in comments to authors below.

We thank the reviewer for their positive assessment and their constructive feedback on the discussion. Also in light of the other reviewers we have made a sincere effort to shorten, restructure and improve the discussion. Additionally, we have addressed all the specific comments the reviewer had below. We appended each change to the manuscript where appropriate below and have addressed all comments in the main text. Having that said, we see this paper and discussion to provide our most up-to-date and personal perspective on a correct concept on the interplay of beta oscillations and dopamine that is generalizable. Providing a concept that is so generalizable is very challenging and so far very few authors have even attempted this. One notable exception is the “status quo” concept by Fries & Engel. While we will do our very best to address the comments, we have decided not to deviate from our initial ambition to provide a discussion on a generalizable concept. Naturally such a concept must be very complex and therefore it will be hard to understand in parts. Through the revision, we hope that the readability and comprehensibility has improved, while it provides an in-depth perspective and hypothesis on how beta oscillations, dopamine and their brain circuits may facilitate brain function. Nevertheless, we want to express our honest gratitude for the thoroughness with which the reviewer has read and scrutinized our paper. The review clearly tells that the reviewer had the ambition to follow and understand what we were trying to convey, which can be rare nowadays. We are truly thankful for this.

The first sentence is not quite true, as invasive neurophysiology was not, and cannot be, done in healthy humans. "The present study combined three openly available datasets of invasive neurophysiology, MRI connectomics, and molecular neuroimaging in healthy humans to characterise the spatial distribution of brain regions exhibiting resting beta activity, their shared circuit architecture, and its correlation with molecular markers of dopamine signaling in the human brain."

Changes in manuscript: We have now removed the “healthy” from the respective sentence.

"Our results motivate to conceptualise the capacity to generate.... This is not clear.

Changes in manuscript: “Our results suggest that one common denominator of brain regions that generate beta activity, is their affiliation with beta oscillations as a feature that arises from a largescale global brain network that is modulated by dopamine.”

"Similarly, the robust beta modulation that is elicited by voluntary action in sensorimotor cortex and its correlation with motor symptoms of Parkinson's disease is long known" - the association between movement-related cortical beta desynchronization and Parkinson's motor signs is not well described - could the authors specify and reference this?

We thank the reviewer for pointing out this lack of clarity. We meant that independently beta is known for “movement” and for “movement disorders” and not “movement in movement disorders”. Having that said, there are some studies that suggest that beta ERD is altered in PD (e.g. <https://doi.org/10.1093/cercor/bht121>), but saying that this is “long known” would be an overstatement and was not our intention. We rephrased this sentence accordingly.

Changes in manuscript: The sentence now reads: “Moreover, the robust beta modulation that is elicited by voluntary action in sensorimotor cortex and its correlation with motor symptoms of Parkinson’s disease is long known.”

“...first fast-cyclic voltammetry experiments that allowed for combined measurement of dopamine release with invasive neurophysiology have provided first evidence that beta band oscillations in healthy non-human primates can differentially link dopamine release, beta oscillations and reward and motor control, depending on the contextual information and striatal domain” - This is not very clear - not sure what “differentially link” signifies.

I think the fact that this is not easy to understand signifies the complexity that we and the authors of the cited paper from Ann Graybiel’s lab aimed to communicate. In fact, we stayed very close to the phrasing used in their paper to try and avoid confusion (Title: Dopamine and beta-band oscillations differentially link to striatal value and motor control” - <https://doi.org/10.1126/sciadv.abb9226>). The specific results go beyond the scope of the discussion but are very interesting, so I would be happy if our paper would inspire readers to look it up.

Changes in manuscript: We have now adapted the sentence to “In line with this more complex picture, direct measurement of dopamine concentration in non-human primates revealed specific interactions between dopamine release, beta oscillations, reward value and motor control, depending on contextual information and striatal domain. This shows that the relationship of dopamine and beta activity is not solely associated with either reward or movement and depends on where in the striatum beta activity is recorded.”

“In fact, one could argue that it can be contextualised in a recently described framework of neural reinforcement, that serves to orchestrate the re-entrance and refinement of neural population dynamics for the production of neural trajectories” - this is not clear - for example what is a neural trajectory? What is meant by “re-entrance and refinement”?

A neural trajectory refers to the path that the activity of a neural population takes through a high-dimensional space over time. It can be obtained through multivariate analysis of population activity with dimensionality reduction techniques, such as PCA. The concept of low-dimensional representations of high-dimensional neural activity has gained a lot of attention in computational neuroscience ever since high-channel count recordings of neural population activity have become available (an early and prominent example is Churchland et al., 2012 Nature <https://doi.org/10.1038/nature11129>, while a more recent example is Safaie et al., Nature 2023 <https://doi.org/10.1038/s41586-023-06714-0>). The review we refer to by Rui Costa and colleagues (Athalye, V. R., Carmena, J. M. & Costa, R. M. Neural reinforcement: re-entering and refining neural dynamics leading to desirable outcomes. Curr Opin Neurobiol 60, 145–154 (2020) <https://doi.org/10.1016/j.conb.2019.11.023>) suggests that dopamine may serve to modulate the likelihood of a specific pattern to emerge and re-enter the cortex – basal ganglia loop, for the “reliable production of neural trajectories driving skillful behavior on-demand”. We believe that this concept could be revolutionary in our understanding of dopaminergic modulation and disorders and together with colleague Alessia Cavallo have written an invited perspective on this topic (<https://doi.org/10.1111/ejn.16222>), which may help further clarify the topic.

Changes in manuscript: We realize that this aspect may sound a bit unclear or far away from the data in this manuscript. However, given that we have spent more than a decade thinking about beta oscillations and how they can be conceptualized, we would prefer not to entirely change our points and rather bet on the possibility that the concepts become more widely accepted and well-known. Nevertheless, we have now adapted the text to make this a bit more clear:

"We hypothesise that, this "status quo" hypothesis could be equally or maybe even more adequately posed on the neural level. Namely, it could provide insights to what degree a certain activity pattern or synaptic connection is to be strengthened or weakened, in light of neural learning. We propose that this putative function can be contextualised in a recently described framework of neural reinforcement, that serves to orchestrate the re-entrance and refinement of neural population dynamics for the production of neural trajectories."

"....after which it was quickly translated to first experimental studies using cortical or subcortical beta signals in human patients⁴⁴." - reference 44 only deals with the use of subcortical beta, not cortical, in adaptive control.

The reviewer is right, in fact there is no study using motor cortex beta for adaptive DBS yet, but different studies have used different markers (especially gamma) since then.

Changes in manuscript: We have rephrased and added citations accordingly: "This approach, also termed adaptive DBS, was first demonstrated based on cortical beta activity that was used to adapt pallidal DBS in the MPTP non-human primate model of PD⁴³. It was quickly translated to first experimental studies using subcortical beta signals in human patients⁴⁴, followed by further research using more complex cortical and subcortical sensing setups and biomarker combinations^{45,46}."

The paragraph headed " Implications for neurotechnology" is quite long and should be condensed and focused. It doesn't seem to support the last sentence, "....targeted interventions that can increase and decrease beta activity, as recently shown through phase specific modulation⁴⁵ could be utilised to mimic phasic dopamine release as a neuroprosthetic approach to alter neural reinforcement³⁸." - I don't quite follow the logic. The authors have clearly shown that beta-related circuits tend to be those linked to dopamine modulation, and may subserve tasks for which reinforcement learning is an important mechanism. However the logic of how modulation of beta activity can "substitute" for modulation of dopamine isn't clear. That would seem to require that the mechanism by which dopamine produces reinforcement, is via an effect on beta oscillation properties (phase, amplitude, frequency). Is there evidence for this? If so it should be better spelled out.

We realize that this is very speculative at this point. Indeed, we believe that subthalamic DBS can mimic dopaminergic control and in the future there may be new treatment avenues, e.g. using neurochemical using neurochemical interfaces for which beta could be informative to mimic dopamine release but ultimately explaining this would be very complex, so we have removed the sentence. With regard to the remaining text in the section, we considered shortening / condensing but felt that this paragraph is highly relevant for the ongoing development of neurotechnology and therefore decided to only remove the first and last sentences.

Changes in manuscript: We have removed the first and last sentences.

"While the abovementioned prospects are promising we should cautiously consider the limitations of our study." - an unnecessary sentence to start a "limitations" section, its

clearly a paragraph about limitations. In general, authors should go thru discussion and reduce verbosity; it is not nearly as well edited as the rest of the paper.

Agreed.

Changes in manuscript: We removed the sentence.

Reviewer #3 (Public Review):

Summary:

In this paper, Chikermane et al. leverages a large open dataset of intracranial recordings (sEEG or ECoG) to analyze resting state (eyes closed) oscillatory activity from a variety of human brain areas. The authors identify a dominant proportion of channels in which beta band activity (12-30Hz) is most prominent and subsequently seek to relate this to anatomical connectivity data by using the sEEG/ECoG electrodes as seeds in a large set of MRI data from the human connectome project. This reveals separate regions and white matter tracts for alpha (primarily occipital) and beta (prefrontal cortex and basal ganglia) oscillations. Finally, using a third available dataset of PET imaging, the authors relate the parcellated signals to dopamine signaling as estimated by spatial uptake patterns of dopamine, and reveal a significant correlation between the functional connectivity maps and the dopamine reuptake maps, suggesting a functional relationship between the two.

Strengths:

Overall, I found the paper well justified, focused on an important topic, and interesting. The authors' use of 3 different open datasets was creative and informative, and it significantly adds to our understanding of different oscillatory networks in the human brain, and their more elusive relation with neuromodulator signaling networks by adding to our knowledge of the association between beta oscillations and dopamine signaling. Even my main comments about the lack of a theta network analysis and discussion points are relatively minor, and I believe this paper is valuable and informative.

Weaknesses:

The analyses were adequate, and the authors cleverly leveraged these different datasets to build an interesting story. The main aspect I found missing (in addition to some discussion items, see below) was an examination of the theta network. Theta oscillations have been involved in a number of cognitive processes including spatial navigation and memory, and have been proposed to have different potential originating brain regions, and it would be informative to see how their anatomical networks (e.g. as in Figure 2) look like under the author's analyses.

The authors devote a significant portion of the discussion to relating their findings to a popular hypothesis for the function of beta oscillations, the maintenance of the "status quo", mostly in the context of motor control. As the authors acknowledge, given the static nature of the data and lack of behavior, this interpretation remains largely speculative and I found it a bit too far-reaching given the data shown in the paper. In contrast, I missed a more detailed discussion on the growing literature indicating a role for beta in mood (e.g. in Kirkby et al. 2018), especially given the apparent lack of hippocampal and amygdala involvement in the paper, which was surprising.

We thank the reviewer for their insightful review of our manuscript. One of the aims of our paper was to provide the ground for a circuit-based conceptualization of beta activity, which does not primarily relate to behavior. Practically we have the ambition to provide a

generalizable concept that can be applied to all behavioral domains including mood. The reason we focus on the “status quo” hypothesis, is that it is one of the very few if not only generalizable concept of the function of beta oscillations. Through our paper and the discussion, we have to redirect this concept towards a less cognitive/behavioral and more anatomical network based domain, while acknowledging principles that may overlap. We realize that this is very ambitious and this endeavour is necessarily very complex and not easy to communicate. In light of the reviewers comments, we have made an effort to improve the discussion as best we could without trailing too far away from what our initial aim was. We are thankful for the suggested reference, which we have now added to the discussion in the section where we have previously discussed beta as biomarker for mood, also noting the absence of beta dominant channels in amygdala and hippocampus. Here it should be clarified however, that a) only three channels were located in the amygdala of which one exhibited beta activity, we should be cautious to not overinterpret this result and b) most channels exhibited beta and just because beta wasn't dominant, it doesn't mean that beta is not present or important in these brain areas. Absence of evidence is not evidence for absence with the way we approached the analysis. We are thankful for the interesting reference, which we have now included our discussion. Notably the study used a complex network analysis, which we could not perform because we did not have parallel recordings from these areas in multiple patients. This is now noted in the limitations.

Changes in manuscript: “For example, it was shown that beta is implicated in working memory²⁸, utilisation of salient sensory cues²⁹, language processing³⁰, motivation³¹, sleep³², emotion recognition³³, mood³⁴ and may even serve as a biomarker for depressive symptom severity in the anterior cingulate cortex³⁵” and “One impactful study reported that beta oscillatory sub-networks of Amygdala and hippocampus could reflect human variations in mood ³⁴. This is interesting, but highlights another relevant limitation of our study, namely that recordings in different areas were stemming from different patients and thus, such sub-network analyses on the oscillatory level could not be conducted.”

Major comment:

• *Although the proportion of electrodes with theta-dominant oscillations was lower (~15%) than alpha (~22%) or beta (~57%), it would be very valuable to also see the same analyses the authors carried out in these frequency bands extended to theta oscillations.*

We agree with the reviewer and appreciate the interest in other frequency bands; theta, alpha and gamma. Our primary interest was to provide a network concept of beta activity, but anticipated that interest would go beyond that frequency band. However, we also had to limit ourselves to what is communicable and comprehensible. The key aim for us was to provide a data-driven circuit description of beta activity that can lay ground for a generalizable concept of where beta oscillations emerge. Reproducing all analyses for every frequency band would clutter both the results and the discussion. Moreover, the honest truth is that funding and individual career plans of the researchers currently do not allow to allocate time for a reanalysis of all data which would be a significant effort. Therefore, we have decided to just add the topography of theta and gamma channels as a supplement. In case the reviewer is interested on a collaboration on extending this project to other frequency bands and circuits, we would like to invite them to get in touch and perhaps this could be a new collaborative project. Until then, we have extended our limitation that this would be important work for the future.

Changes in manuscript:

We have added and cited the new supplementary figure for the results from theta in the results section, which now reads:

“Further information on the topography of theta channels are shown in supplementary figure 1.”

We would like to add that a sensible interpretation of results from gamma dominant channels is unlikely to be possible given the low count of channels with prominent resting activity in this frequency band. We have added the following text to the limitations section: “The aim of this study was to elucidate the circuit architecture of beta oscillations, which is why insights from this study for other frequency bands are limited. Future research investigating the specific circuits of theta, alpha and gamma oscillations and their relationship with neurotransmitter uptake could yield new important insights on the networks underlying human brain rhythms.”

Reviewer #3 (Recommendations For The Authors):

Minor comments:

• *Results: "we performed non-parametric Spearman's correlations between the structural and functional connectivity maps of beta networks with neurotransmitter uptake". This is a significantly complex analysis that requires more detail for the reader to evaluate. There is more detail in the Figure 3 legend but still insufficient. The Methods offer more detail, but I found the description of the parcellation to be vague and I would appreciate a more detailed description.*

We thank the reviewer for bringing the insufficient explanation of the methods used to calculate the correlations in analysis to our attention. We have now made an effort to provide more level of detail in the relevant paragraphs.

Changes in manuscript: We have now made changes to both the Results and Methods sections and added the following explanations respectively:

Results: “Next, we resliced the beta network map and the PET images to allow for a meaningful comparison, using a combined parcellation with 476 brain regions that include cortex19, basal ganglia20, and cerebellum21. Here, each parcel – which was a collection of voxels belonging to a particular brain region – from the connectivity map was correlated with the same parcel containing average neurotransmitter uptake from the respective PET scan (see Figure 3A). In this way nonparametric Spearman’s correlations between PET intensity and structural and functional connectivity maps of beta networks were obtained, which indicate to what degree the spatial distribution of connectivity is similar to the distribution of neurotransmitter uptake.”

Methods: “A custom master parcellation in MNI space was created in Matlab using SPM functions by combining three existing parcellations to include cortical regions19, structures of the basal ganglia20 and cerebellar regions21. Regions that were (partially) overlapping between the atlases were only selected once. The final compound parcellation had 476 regions in total. This parcellation was applied to both PET and structural and functional connectivity maps using SPM and custom code. This allowed for the calculation of spatial correlations, providing a statistical measure of spatial similarity of the PET intensity and MRI connectivity distributions. For this, Spearman’s ranked correlations were used to calculate correlations between the PET images, such as the dopamine aggregate map and both functional and structural beta connectivity networks (Figure 3). The analysis was repeated for individual tracers showing similar results Supplementary figure 2. Finally, to validate these results, a control analysis was performed using a GABA PET scan from the same open dataset of neurotransmitter uptake following the same pipeline (Figure 2A, 2B).”

- *All of the recordings were taken in an eyes-closed condition. This is likely to affect the power of alpha oscillations; the authors should comment on this.*

We agree with the reviewer that this will likely have influenced the results. However, given that the key result of our paper is the abundance and circuit topography of beta oscillations, it is unlikely that increased alpha in some channels will have led to false positive results for beta. If anything, it may have increased the contrast leading to a more conservative estimate of which channels truly show strong beta dominance. On the other hand, we should acknowledge that this limitation can affect the interpretation of the alpha result. Another reason for us to primarily focus on beta in the discussion and results presentation.

Changes in manuscript: We now comment on this in the results:

“It should be noted that that alpha recordings were performed in eyes closed which is known to increase alpha power, which may influence the generalizability of the alpha maps to an eyes open condition. However, given that our primary use of alpha was to act as a control, we believe that this should not affect the interpretability of the key findings of our study.”

- *Although the relative proportion of theta and gamma channels is lower, it would be interesting to see the distribution of channels in a SOM figure.*

As described above, we have now added supplementary figure 1 that accommodates the topography but not the network analyses.

- *Figure legend - typo - "Neither, alpha nor beta" - no comma needed.*

Now fixed, thank you for pointing is to this lapse!

- *Results: " ere, we aimed to investigate the whole brain circuit representation of beta activity, which is impossible with current neurophysiology approaches" not entirely accurate; suggest rephrasing it to "Here, we aimed to investigate the whole brain circuit representation of beta activity, which is impossible with non-invasive neurophysiology approaches "*

Thank you for suggesting the alternative formulation.

Changes in manuscript: The text has been modified as per the suggestion and now reads “Here, we aimed to investigate the whole brain circuit representation of beta activity, which is impossible with non-invasive neurophysiology approaches”.

- *Results - typo - "cortical brain areas, that exhibit resting beta activity share a common brain network" - no comma needed.*

Thank you for the suggestion, the comma has been removed to better the flow of the sentence structure as suggested.

<https://doi.org/10.7554/eLife.97184.2.sa0>